

# ATOSFleetManagement: Architecture and Development Guide

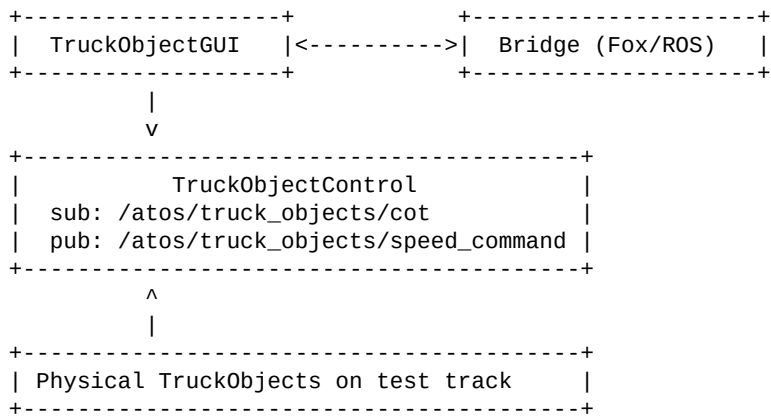
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## 1) Scope

ATOSFleetManagement is a lightweight ATOS variant for TruckObject development. This stack intentionally removes scenario engine and ISO22133 dependencies from runtime behavior.

- Included: TruckObjectControl, TruckObjectGUI, bridge stack
- Excluded: OpenScenarioGateway, JournalControl, EsminiAdapter, ObjectControl, ISO22133 runtime path

## 2) Runtime Graph



## 3) Placeholder Control Logic

Input COT topic: /atos/truck\_objects/cot (std\_msgs/String)

Expected payload: id=<truck\_id>;distance\_m=<value>;tcp\_connected=<0|1>

Policy:

- Consider only trucks with valid TCP flag and fresh COT
- Compute minimum distance gap along trajectory among eligible trucks
- If gap < 200 m: publish target\_speed\_kmh=0
- Else if gap < 400 m: publish target\_speed\_kmh=30
- Else: no new speed restriction command

Output topic: /atos/truck\_objects/speed\_command (std\_msgs/String)

## 4) Build and Run

```
source /opt/ros/humble/setup.bash
cd /home/sepast/Documents/repos/ATOS
./scripts/build_atosfleetmanagement.sh
```

```
source /opt/ros/humble/setup.bash
source /home/sepast/atos_ws/install/setup.bash
ros2 launch atos launch_atosfleetmanagement.py insecure:=True
```